Regression Channel

Indicators & Analysis / Trend | intermediate | 5 min read | DeepCharts Help Center | August 24, 2026

The Regression Channel is a tool used to identify trends and areas of support and resistance. It draws a channel derived from linear regression: a central trend line fitted through price, flanked by upper and lower bands offset by a chosen number of standard deviations.

Where the Linear Regression indicator plots a single rolling line, the Regression Channel frames the whole move — showing the trend's direction, its statistical center, and the zone where most price action has stayed.

What it is

The Regression Channel answers the question: what is the current trend, and how far from it is price allowed to wander before that means something? The center line is the best-fit regression through the lookback window; the bands sit a configurable number of standard deviations above and below it. Price near the center line is at the trend's fair value; price at a band is statistically stretched, favoring either a reversion toward the center or — if it breaks and holds — a change in trend.

[SCREENSHOT PLACEHOLDER] A candlestick chart with the Regression Channel applied — upward-sloping center line with upper and lower deviation bands containing the trend, price tagging the lower band and reverting to the center

`dc-en-regression-channel-01.png`

When to use it

- Framing a trend objectively: the channel's slope states direction, its width states volatility.
- Trading pullbacks within a trend — entries near the channel's lower band in an up-slope, exits or shorts near the upper band in a down-slope.
- Detecting trend breaks: price closing and holding outside the channel suggests the regression that defined the move no longer fits.
- Pairing with swing structure via the Zig Zag mode so the channel anchors to detected swings rather than a fixed bar count.

Quick start

1. Open a chart — see Open Your First Chart.
2. Click the bar-chart icon in the chart's top-left icon row to open the **Indicators** panel, then click **Indicators** to open the full **Indicator List**.
3. Search for **Regression Channel** and click **+** to add it.
4. Click the gear icon to open its settings.
5. Starter configuration: keep **Bars** at the default 100 and **Value Dev. Std.** at the default 1.00. Widen the deviation (for example toward 2.00) if you want the bands to contain more of the price action and flag only larger stretches.

[SCREENSHOT PLACEHOLDER] Regression Channel settings dialog showing the General section with Mode, Value Dev. Std. = 1.00 and Bars = 100, plus the Zig Zag section and the MID/UP/DN plot sections below

`dc-en-regression-channel-02.png`

How to read it

- **Center line slope** — the trend. Up-slope, favor longs; down-slope, favor shorts; near-flat, treat the channel as a range.

• **Position within the channel** — price near the center line is at the fitted trend's fair value; price at a band is at a statistical extreme for the window.
• **Band touches** — in a healthy trend, touches of the band on the trend side (lower band in an uptrend) are pullback opportunities; repeated failures to reach the far band show the trend weakening.
• **Channel breaks** — a decisive close outside the channel that holds means the fitted trend is failing; expect either acceleration in the break direction or a new regime the channel will re-fit to.
• **Trend coloring** — the MID, UP and DN lines each have positive and negative colors, so the channel's own coloring tells you which direction the fit currently has.

Settings reference

General

Setting	What it does
Mode	Selects how the channel operates. **[CONFIRM: the exact list of Mode options and what each does — the live article names the setting without enumerating its values]**
Value Dev. Std.	Number of standard deviations for the upper and lower bands. Default: 1.00. Higher values place the bands further from the center line.
Bars	Number of bars used for the regression calculation. Default: 100.

Zig Zag

Setting	What it does
Zig Zag Mode	Enables swing-pattern detection so the channel works from Zig Zag swings.
Zig Zag Abs.Rev.	Tick reversal threshold for swing detection. Default: 0.50.
Zig Zag Tick Rev./Highest-Lowest	Reversal value for the swing calculation. Default: 22.

MID — center line plot

Setting	What it does
Line Width	Center line thickness. Default: 2.
Line Style	Solid, Dashed, Dotted and other formats.
Positive Color	Color when the fitted trend is up.
Negative Color	Color when the fitted trend is down.

UP — upper line plot

Setting	What it does
Line Width	Upper band thickness. Default: 2.
Line Style	Solid, Dashed, Dotted and other formats.
Positive / Negative Color	Trend-based coloring for the upper band.

DN — lower line plot

Setting	What it does
Line Width	Lower band thickness. Default: 2.
Line Style	Available line format options.
Positive / Negative Color	Trend-based coloring for the lower band.

Tips and common mistakes

- **Match Bars to the move you are framing.** 100 bars fitted over two separate trends produces a meaningless average channel; shorten the window, or use **Zig Zag Mode** to anchor the channel to detected swings.
- **One standard deviation contains less than you may expect.** At the default 1.00, price will trade outside the bands regularly in normal conditions — that alone is not a signal. Raise **Value Dev. Std.** if you want band touches to be rarer, more meaningful events.
- **Do not fade every band touch.** In a strong trend, price can ride the far band for extended stretches; check the center line's slope before treating a band as a reversal level.
- Compare with Zig Zag to understand the swing detection that **Zig Zag Mode** builds on, and with Standard Deviation for the dispersion measure behind the bands.

Related articles

- Linear Regression
- Zig Zag
- Standard Deviation
- Bollinger Bands
- Different Types of Input Data for Indicators
- How to Change the Layout of Indicators